



The Earthquake — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Chapter 05_The_Earthquake

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Plate Tectonics Theory — Origin

- Plate Tectonics Theory H.H. Hess ne akela 1967 mein NAHI diya (Wilson, Morgan, Parker, McKenzie, Hess sabne mila ke 1960s late mein develop kiya) — Statement I **FALSE**; Theory Sea-floor Spreading pe based hai — Statement II **TRUE**. [Q1]

Seismic Waves — Types

- Earthquake se **P-waves, S-waves, AND Surface waves** teeno generate hote hain — P-waves fastest (compressional), S-waves liquid se pass nahi hoti (shear), Surface waves (Love + Rayleigh) baad mein aati hain. [Q2]
- **Shadow Zone** — har earthquake ka apna alag shadow-zone pattern hota hai (Statement I TRUE); lekin 105° se aage seismometers dono P aur S waves record NAHI karte (P-waves $105-140^\circ$ ke beech missing, S-waves apne shadow zone se aage missing) — Statement II **FALSE**. [Q3]

Ring of Fire

- **'Ring of Fire'** — Earthquakes, Volcanoes, aur Pacific Ocean teeno se associated hai (world ke ~90% earthquakes yahin aate hain) — Forest fires se related NAHI hai. [Q4]

Richter Scale — Logarithmic Scale

- Richter Scale **logarithmic (base 10)** hai — 1 magnitude unit increase = **10x amplitude** (100x amplitude NAHI); energy release ~**32x** badhta hai per unit (100x nahi). [Q5, Q6]
- Charles Francis Richter ne **1935** mein Richter scale develop kiya. Magnitude = energy release ka measurement, Intensity = **Mercalli Scale** pe measure hoti hai. [Q7, Q8]

⚠️ EXAM TRAP

- Richter scale mein 1-unit increase = **10x amplitude**, **~32x energy** — '100x' wala option common trap hai, sahi nahi hai. [Q5, Q6]

Instruments — Seismometer/Seismograph

- **Seismometer** earthquake measure karta hai (sensing component) — heartbeat/tree-height se koi relation nahi. [Q9]
- **Seismograph** seismic waves detect+record karne ka instrument/system hai; **Seismogram** = recorded trace/output hai. [Q10]

Tsunami — Origin aur Major Events

- 'Tsunami' word **Japanese** hai (tsu=harbour, nami=wave, literal meaning 'harbour wave'). [Q11]
- 2004 Indian Ocean Tsunami: Sumatra (Indonesia) ke paas earthquake, India-Burma plate boundary pe ~1,300km fault slip ~15m, Africa (Somalia) tak effect pahuncha, Bangladesh mein kam casualties hui (fault orientation ki wajah se). Death toll **~2-3 lakh tha, 50 lakh NAHI** (yeh galat statement hai). [Q12]
- 2011 Japan Tsunami ne **Fukushima** region ko severely affect kiya (9.0 magnitude earthquake, Fukushima Daiichi Nuclear Plant accident). [Q13]

⚠️ EXAM TRAP

- 2004 Tsunami death toll **~2-3 lakh** tha — '50 lakh se zyada' wala claim EXAM key mein GALAT statement hai, isse confuse mat ho. [Q12]

India ka NW Region — Earthquake Prone

- India ka North-Western region earthquake-prone hai kyunki **Plate Tectonic Activity** (Indian Plate-Eurasian Plate convergence zone) — volcanic activity ya coral formation ki wajah se NAHI. [Q14]

Focus vs Epicentre

- **Focus/Hypocentre** = Earth ke andar wo point jahan earthquake energy release hoti hai; **Epicentre** = Focus ke exactly upar surface ka point — seismic waves surface pe sabse pehle epicentre ke paas pahunchti hain. [Q15]

Plate Tectonics — Cocos Plate

- **Cocos Plate** Central America aur Pacific Plate ke beech located hai. [Q16]